

Paul Tiede

Black Hole Initiative at Harvard University
20 Garden Street Cambridge MA, 02138
✉ ptiede@fas.harvard.edu
🌐 ptiede.github.io

Curriculum Vitae

Research Interests

Data science, Bayesian inference, scientific computing, probabilistic programming, computational imaging, machine learning, black holes

Research Experience

- 2025– **Black Hole Initiative at Harvard University**, *Research Associate*
BHEX chief data scientist
- 2024–2025 **Center for Astrophysics | Harvard & Smithsonian**, **Black Hole Initiative**, *BHEX Fellow*
- 2021–2024 **Center for Astrophysics | Harvard & Smithsonian**, *Postdoctoral Researcher*
- 2021–2022 **Black Hole Initiative**, *BHI Postdoctoral Fellow*
- 2017–2021 **Department of Physics and Astronomy, University of Waterloo**, *Graduate Research Assistant*
- 2015–2017 **Department of Applied Mathematics, University of Waterloo**, *Research Assistant (Master's)*
- 2012–2015 **Department of Physics and Astronomy, University of Waterloo**, *Undergraduate Research Assistant*

Education

- 2017–2021 **University of Waterloo and Perimeter Institute**, *Ph.D. Physics*, Waterloo, ON
- 2015–2017 **University of Waterloo**, *MMath Applied Mathematics*, Waterloo, ON
- 2010–2015 **University of Waterloo**, *B.Sc Mathematical Physics*, *Specialization: Astrophysics*, *Dean's Honors List*, Waterloo, ON

Awards & Honours

- 2025 **Event Horizon Telescope Collaboration**, *Early Career Award*
Awarded for leadership in the EHT collaboration project on the multiyear polarized variability of M87*
- 2022 **Event Horizon Telescope Collaboration**, *Early Career Award*
Awarded for seminal contributions to developing innovative modeling frameworks to allow precise measurements from EHT observations of Sgr A*
- 2021 **Event Horizon Telescope Collaboration**, *Outstanding Thesis Award*
For his many contributions to EHT data analysis tools, particularly with respect to the study of variability
- 2020 **AAS High Energy Group (Shared with EHT Collaboration)**, *Bruno Rossi Prize*
- 2019 **Fundamental Physics Prize Foundation (Shared with EHT Collaboration)**, *Breakthrough Prize in Fundamental Physics*
- 2017–2020 **National Science Engineering Research Council (NSERC)**, *Alexander Graham Bell Scholarship (CGS-D)*
Value: \$ 35,000 per year
- 2017–2020 **University of Waterloo**, *President's Graduate Scholarship (Doctorate)*
Value: \$ 10,000 per year
- 2017 **University of Waterloo**, *Applied Mathematics Outstanding Teacher Assistant Award*
- 2015–2016 **NSERC**, *Canada Graduate Scholarship (CSGS-M)*
Value: \$ 17,500 per year
- 2015–2016 **University of Waterloo**, *President's Graduate Scholarship (Masters)*
Value: \$ 10,000 per year
- 2015 **NSERC**, *Undergraduate Student Research Award*

- 2014 **University of Waterloo, Helen Sawyer Hogg Scholarship**
Value: \$ 500
- 2013 **NSERC, Undergraduate Student Research Award**
- 2012 **NSERC, Undergraduate Student Research Award**
- 2010 **President's Scholarship, University of Waterloo**
Value: \$ 2,000

Grants Awarded

- 2025–2027 **Harvard Data Science Initiative & Amazon Web Services, AWS Impact Computing Project**
Value: \$599,245

Successful Observing Proposals

- 2025–2026 **National Radio Astronomy Observatory, VLBA Observing Proposal**
96 hours of VLBA, Greenland telescope, and Haystack 37m time to monitor M87* at 43 & 86 GHz

Outreach

- 2026 **Public Talk at the EcoTarium in Worcester MA, *Imaging Black Holes***
I was asked to give a public outreach talk in February 2026 to the local EcoTarium science center.
- 2023 **Sq'ewquel Community School, *Imaging Black Holes***
Visited elementary and middle school indigenous students about Black Holes and what it is like being an astrophysicist.
- 2022 **Center for Astrophysics YouTube, *The Quest to Image Black Holes***
Participated in a panel discussion on YouTube about the first image of the black hole Sagittarius A*, with Ellen Stofan, the Under Secretary for Science and Research at The Smithsonian.
- 2015–2017 **Waterloo ON, *Let's Talk Science***
 - Youth science education program focused on increasing STEM literacy and improving science education.
 - Helped develop a physics lab day for high school students to learn about a variety of physics disciplines.

Software

GitHub userid: **ptiede**

Primary/Core Developer

Comrade.jl	Julia	https://github.com/ptiede/Comrade.jl
VIDA.jl	Julia	https://github.com/ptiede/VIDA.jl
EHTJulia Org	Julia	https://github.com/EHTJulia
THEMIS	C++ & MPI	(Private repository)
THEMISPY	Python	https://github.com/aeb/ThemisPy

Academic Service

Reviewer, The Astrophysics Journal
Reviewer, Astronomy & Astrophysics
Reviewer, Galaxies MDPI
Reviewer, Physical Review Journal
Reviewer, Monthly Notices of the Royal Astronomical Society

Presentations

- 2026 **JuliaCon 2026 Keynote speaker, *Computational Black Hole Data Science with Julia***, (Invited)
- 2025 **SCEECS Seminar, *Horizon Scale Polarized Variability of M87* from 2017–2021***, (Invited)

- 2025 **University of Amsterdam**, *Astronomy Seminar*, Horizon Scale Polarized Variability of M87* from 2017–2021 (Invited)
- 2025 **CANSSI Monte Carlo Workshop**, *Monte Carlo Applications in Radio Astronomy*, (Invited)
- 2025 **JuliaCon 2025**, *Building an Astronomy Code for VLBI in Julia*, (Organizer)
- 2025 **Academia Sinica Institute of Astronomy and Astrophysics**, *Interpol: A Polarized Imaging and Calibration Code for Heterogeneous VLBI Arrays*, (Invited)
- 2025 **Academia Sinica Institute of Astronomy and Astrophysics**, *Comrade VLBI Analysis Workshop*, (Invited)
- 2025 **EHT Collaboration Meeting: Berlin**, *M87* Multiyear Collaboration Project*, (Co-Lead, Presenter)
- 2025 **AAS 2025 (Winter)**, *Evolution of the Ring Image of M87 over Four Years of Observation*, (Invited)
- 2024 **RMS Seminar Harvard & Smithsonian**, *Automated Bayesian Black Hole Imaging for Next Generation Science*, (Invited)
- 2024 **Harvard Data Science Institute Seminar**, *Accelerating Black Hole Imaging with Computational Data Science*, (Invited)
- 2024 **AstroAI Seminar Series**, *Accelerating Bayesian Imaging of Black Holes With Julia*, (Invited)
- 2024 **MIT CSE Community Seminar**, *Unlocking Black Hole Physics with Computational Astronomy*, (Invited)
- 2024 **Black Holes Inside and Out 24**, *On The Photon Ring in M87* Paper & Nature of Object*, (Invited)
- 2024 **AI for Radioastronomy**, *Automated Bayesian Polarized Imaging and Calibration for Heterogeneous VLBI and Black Hole Imaging*, (Invited)
- 2023 **JuliaCon 2023**, *Computational Radio Astronomy with Julia Mini-symposium*, Organizer and Speaker
- 2023 **EAS 2023**, *Bayesian Inference Algorithms for Imaging and Modeling with the EHT & ngEHT*, (Invited)
- 2023 **Astrophysical Black Holes: A Rapidly Moving Field**, *A Black Hole Closeup: Enhancing Black Hole Science with the ngEHT and Space*, (Invited)
- 2023 **UBC Statistics Seminar**, *Statistical Imaging of Black Holes Using the EHT*, (Invited)
- 2023 **Enzyme Conference 2023**, *Accelerating Black Hole Imaging with Enzyme*, (Invited)
- 2022 **Black Hole Initiative Conference 2022**, *Measuring the Variability, Morphology and Mass of Sgr A**, (Invited)
- 2022 **NERQUAM 2022**, *Modeling and Interpreting the Sgr A* Results*, (Invited)
- 2021 **Black Hole Initiative (BHI) Harvard University**, *Variational Image Domain Feature Extraction and Modeling the Horizon*, (Invited)
- 2019 **Black Hole Initiative (BHI) Harvard University**, *Spacetime Tomography with the Event Horizon Telescope*, (Invited)

Publications (ORCID ID: 0000-0003-3826-5648)

First Author and Primary Collaboration Contribution

- [1] **Paul Tiede**, William Moses, Valentin Churavy, Michael D. Johnson, Dominic Pesce, Lindy Blackburn, and Peter Galison. Hierarchical Interferometric Bayesian Imaging. *arXiv e-prints*, page arXiv:2511.17706, November 2025.
- [2] Event Horizon Telescope Collaboration. (**paper co-lead**) Horizon-scale variability of M87* from 2017–2021 EHT observations. *arXiv e-prints (accepted to A&A)*, page arXiv:2509.24593, September 2025.

- [3] **Paul Tiede**. Very Long Baseline Interferometry as a Precision Strong Gravity Instrument. *General Relativity and Gravitation*, 57(6):98, June 2025.
- [4] Event Horizon Telescope Collaboration. **(analysis)** The persistent shadow of the supermassive black hole of M87: II. Model comparisons and theoretical interpretations. *Astronomy and Astrophysics*, 693:A265, January 2025.
- [5] **Paul Tiede** and Avery E. Broderick. Bayesian Constraints on the Ring Ellipticity of M87* 2017 Using THEMIS. *ApJ*, 971(2):172, August 2024.
- [6] Event Horizon Telescope Collaboration. **(analysis)** First Sagittarius A* Event Horizon Telescope Results. VII. Polarization of the Ring. *ApJL*, 964(2):L25, April 2024.
- [7] Event Horizon Telescope Collaboration. **(paper writing and analysis)** The persistent shadow of the supermassive black hole of M 87. I. Observations, calibration, imaging, and analysis. *A&A*, 681:A79, January 2024.
- [8] **Paul Tiede**, Avery E. Broderick, Daniel C. M. Palumbo, and Andrew Chael. Measuring the Ellipticity of M87* Images. *ApJ*, 940(2):182, December 2022.
- [9] **Paul Tiede**, Michael D. Johnson, Dominic W. Pesce, Daniel C. M. Palumbo, Dominic O. Chang, and Peter Galison. Measuring Photon Rings with the ngEHT. *Galaxies*, 10(6):111, December 2022.
- [10] Event Horizon Telescope Collaboration. **(paper writing and analysis)** First Sagittarius A* Event Horizon Telescope Results. III. Imaging of the Galactic Center Supermassive Black Hole. *ApJL*, 930(2):L14, May 2022.
- [11] Event Horizon Telescope Collaboration and **(paper writing and analysis team)**. First Sagittarius A* Event Horizon Telescope Results. IV. Variability, Morphology, and Black Hole Mass. *ApJL*, 930(2):L15, May 2022.
- [12] Event Horizon Telescope Collaboration. **(paper analysis)** First Sagittarius A* Event Horizon Telescope Results. V. Testing Astrophysical Models of the Galactic Center Black Hole. *ApJL*, 930(2):L16, May 2022.
- [13] **Paul Tiede**, Avery E. Broderick, and Daniel C. M. Palumbo. Variational Image Feature Extraction for the Event Horizon Telescope. *ApJ*, 925(2):122, February 2022.
- [14] Event Horizon Telescope Collaboration. **(paper analysis)** First Sagittarius A* Event Horizon Telescope Results. VI. Testing the Black Hole Metric. *ApJL*, 930(2):L17, May 2022.
- [15] **Paul Tiede**, Hung-Yi Pu, Avery E. Broderick, Roman Gold, Mansour Karami, and Jorge A. Preciado-López. Spacetime Tomography Using the Event Horizon Telescope. *ApJ*, 892(2):132, April 2020.
- [16] Event Horizon Telescope Collaboration. **(paper writing and analysis)** first M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole. *ApJL*, 875(1):L6, April 2019. Publisher: IOP Publishing.
- [17] **Paul Tiede**, Avery E. Broderick, Mohamad Shalaby, Christoph Pfrommer, Ewald Puchwein, Philip Chang, and Astrid Lamberts. Constraints on the Intergalactic Magnetic Field from Bow Ties in the Gamma-Ray Sky. *ApJ*, 892(2):123, April 2020.
- [18] **Paul Tiede**, Avery E. Broderick, Mohamad Shalaby, Christoph Pfrommer, Ewald Puchwein, Philip Chang, and Astrid Lamberts. Bow Ties in the Sky. II. Searching for Gamma-Ray Halos in the Fermi Sky Using Anisotropy. *ApJ*, 850(2):157, December 2017.

Student Authored Papers

- [1] **Dominic O. Chang**, Michael D. Johnson, and **Paul Tiede**. Assessing the Role of Intrinsic Variability in Black Hole Parameter Inference Using Multiepoch EHT Data. *The Astrophysical Journal Letters*, 989(1):L1, August 2025. doi:10.3847/2041-8213/ade431.

- [2] **Rohan Dahale**, Ilje Cho, Kotaro Moriyama, Kaj Wiik, et al. Origin of the ring ellipticity in the black hole images of M87*. *Astronomy & Astrophysics*, 699:A279, July 2025. doi:10.1051/0004-6361/202555235.
- [3] **Uri Rolls**, Dominic W. Pesce, **Paul Tiede**, Lindy Blackburn, et al. Modeling atmospheric phase corruptions in high-frequency VLBI using Gaussian processes. *arXiv e-prints*, arXiv:2510.19071, October 2025. doi:10.48550/arXiv.2510.19071.
- [4] **Anna Tartaglia**, Kazunori Akiyama, and **Paul Tiede**. ScatteringOptics.jl: An Interstellar Scattering Framework in the Julia Programming Language. *The Journal of Open Source Software*, 10(105):6354, January 2025. doi:10.21105/joss.06354.
- [5] **Dominic O. Chang**, Michael D. Johnson, **Paul Tiede**, and Daniel C. M. Palumbo. Bayesian Black Hole Photogrammetry. *ApJ*, 974(1):143, October 2024. doi:10.3847/1538-4357/ad6b28.
- [6] **Erandi Chavez**, Sara Issaoun, Michael D. Johnson, **Paul Tiede**, et al. Prospects of Detecting a Jet in Sagittarius A* with Very-long-baseline Interferometry. *ApJ*, 974(1):116, October 2024. doi:10.3847/1538-4357/ad6b1f.

Software Papers

- [1] Iniyar Natarajan, Lindy Blackburn, **Paul Tiede**, Dominic Pesce, et al. Anime: Atmospheric and Instrument Models in the Measurement Equation. *The Journal of Open Source Software*, 9(97):6573, May 2024. doi:10.21105/joss.06573.
- [2] Nikola Surjanovic, Miguel Biron-Lattes, **Paul Tiede**, Saifuddin Syed, et al. Pigeons.jl: Distributed Sampling From Intractable Distributions. *arXiv e-prints*, arXiv:2308.09769, August 2023. doi:10.48550/arXiv.2308.09769.
- [3] **Paul Tiede**. Comrade: Composable Modeling of Radio Emission. *The Journal of Open Source Software*, 7(76):4457, August 2022. doi:10.21105/joss.04457.
- [4] Avery E. Broderick, Roman Gold, Mansour Karami, Jorge A. Preciado-López, et al. THEMIS: A Parameter Estimation Framework for the Event Horizon Telescope. *ApJ*, 897(2):139, July 2020. doi:10.3847/1538-4357/ab91a4.

Secondary Author

- [1] Dmitry Ayzenberg, Lindy Blackburn, Richard Brito, Silke Britzen, et al. Fundamental physics opportunities with future ground-based mm/sub-mm VLBI arrays. *Living Reviews in Relativity*, 28(1):4, June 2025. doi:10.1007/s41114-025-00057-0.
- [2] Niayesh Afshordi, Abhay Ashtekar, Enrico Barausse, Emanuele Berti, et al. Black Holes Inside and Out 2024: visions for the future of black hole physics. *arXiv e-prints*, arXiv:2410.14414, October 2024. doi:10.48550/arXiv.2410.14414.
- [3] Michael D. Johnson, Kazunori Akiyama, Rebecca Baturin, Bryan Bilyeu, et al. The Black Hole Explorer: Motivation and Vision. *arXiv e-prints*, arXiv:2406.12917, June 2024. doi:10.48550/arXiv.2406.12917.
- [4] Alexandru Lupasca, Alejandro Cárdenas-Avendaño, Daniel C. M. Palumbo, Michael D. Johnson, et al. The Black Hole Explorer: Photon Ring Science, Detection and Shape Measurement. *arXiv e-prints*, arXiv:2406.09498, June 2024. doi:10.48550/arXiv.2406.09498.
- [5] Dominic W. Pesce, Lindy Blackburn, Ryan Chaves, Sheperd S. Doeleman, et al. Atmospheric Limitations for High-frequency Ground-based Very Long Baseline Interferometry. *ApJ*, 968(2):69, June 2024. doi:10.3847/1538-4357/ad3961.
- [6] Koushik Chatterjee, Andrew Chael, **Paul Tiede**, Yosuke Mizuno, et al. Accretion Flow Morphology in Numerical Simulations of Black Holes from the nGEHT Model Library: The Impact of Radiation Physics. *Galaxies*, 11(2):38, February 2023. doi:10.3390/galaxies11020038.
- [7] Sheperd S. Doeleman, John Barrett, Lindy Blackburn, Katherine L. Bouman, et al. Reference Array and Design Consideration for the Next-Generation Event Horizon Telescope. *Galaxies*, 11(5):107, October 2023. doi:10.3390/galaxies11050107.

- [8] Razieh Emami, Sheperd S. Doeleman, Maciek Wielgus, Dominic Chang, et al. The EB Correlation in Resolved Polarized Images: Connections to the Astrophysics of Black Holes. *ApJ*, 955(1):6, September 2023. doi:10.3847/1538-4357/acdc96.
- [9] Razieh Emami, Angelo Ricarte, George N. Wong, Daniel Palumbo, et al. Unraveling Twisty Linear Polarization Morphologies in Black Hole Images. *ApJ*, 950(1):38, June 2023. doi:10.3847/1538-4357/acc8cd.
- [10] Razieh Emami, **Paul Tiede**, Sheperd S. Doeleman, Freek Roelofs, et al. Tracing Hot Spot Motion in Sagittarius A* Using the Next-Generation Event Horizon Telescope (ngEHT). *Galaxies*, 11(1):23, January 2023. doi:10.3390/galaxies11010023.
- [11] Michael Johnson, Peter Galison, Peter Kurczynski, Tirupati Kumara Sridharan, et al. The Event Horizon Explorer: A New Mission to Detect a Black Hole’s Photon Ring. In *American Astronomical Society Meeting Abstracts*, volume 242 of *American Astronomical Society Meeting Abstracts*, page 330.02. June 2023.
- [12] Michael D. Johnson, Kazunori Akiyama, Lindy Blackburn, Katherine L. Bouman, et al. Key Science Goals for the Next-Generation Event Horizon Telescope. *Galaxies*, 11(3):61, April 2023. doi:10.3390/galaxies11030061.
- [13] Angelo Ricarte, **Paul Tiede**, Razieh Emami, Aditya Tamar, et al. The ngEHT’s Role in Measuring Supermassive Black Hole Spins. *Galaxies*, 11(1):6, January 2023. doi:10.3390/galaxies11010006.
- [14] Freek Roelofs, Lindy Blackburn, Greg Lindahl, Sheperd S. Doeleman, et al. The ngEHT Analysis Challenges. *Galaxies*, 11(1):12, January 2023. doi:10.3390/galaxies11010012.
- [15] Avery E. Broderick, Roman Gold, Boris Georgiev, Dominic W. Pesce, et al. Characterizing and Mitigating Intraday Variability: Reconstructing Source Structure in Accreting Black Holes with mm-VLBI. *ApJL*, 930(2):L21, May 2022. doi:10.3847/2041-8213/ac6584.
- [16] Avery E. Broderick, Dominic W. Pesce, Roman Gold, **Paul Tiede**, et al. The Photon Ring in M87*. *ApJ*, 935(1):61, August 2022. doi:10.3847/1538-4357/ac7c1d.
- [17] Avery E. Broderick, **Paul Tiede**, Dominic W. Pesce, and Roman Gold. Measuring Spin from Relative Photon-ring Sizes. *ApJ*, 927(1):6, March 2022. doi:10.3847/1538-4357/ac4970.
- [18] Sara Issaoun, Maciek Wielgus, Svetlana Jorstad, Thomas P. Krichbaum, et al. Resolving the Inner Parsec of the Blazar J1924-2914 with the Event Horizon Telescope. *ApJ*, 934(2):145, August 2022. doi:10.3847/1538-4357/ac7a40.
- [19] Peter Kurczynski, Michael D. Johnson, Sheperd S. Doeleman, Kari Haworth, et al. The Event Horizon Explorer mission concept. In Laura E. Coyle, Shuji Matsuura, and Marshall D. Perrin, editors, *Space Telescopes and Instrumentation 2022: Optical, Infrared, and Millimeter Wave*, volume 12180 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 121800M. August 2022. doi:10.1117/12.2630313.
- [20] Astrid Lamberts, Ewald Puchwein, Christoph Pfrommer, Philip Chang, et al. Constraining blazar heating with the $2 \lesssim z \lesssim 3$ Lyman- α forest. *MNRAS*, 512(2):3045–3059, May 2022. doi:10.1093/mnras/stac553.
- [21] Daniel C. M. Palumbo, Zachary Gelles, **Paul Tiede**, Dominic O. Chang, et al. Bayesian Accretion Modeling: Axisymmetric Equatorial Emission in the Kerr Spacetime. *ApJ*, 939(2):107, November 2022. doi:10.3847/1538-4357/ac9ab7.
- [22] He Sun, Katherine L. Bouman, **Paul Tiede**, Jason J. Wang, et al. α -deep Probabilistic Inference (α -DPI): Efficient Uncertainty Quantification from Exoplanet Astrometry to Black Hole Feature Extraction. *ApJ*, 932(2):99, June 2022. doi:10.3847/1538-4357/ac6be9.
- [23] Maciek Wielgus, Nicola Marchili, Iván Martí-Vidal, Garrett K. Keating, et al. Millimeter Light Curves of Sagittarius A* Observed during the 2017 Event Horizon Telescope Campaign. *ApJL*, 930(2):L19, May 2022. doi:10.3847/2041-8213/ac6428.

- [24] S. Issaoun, M. D. Johnson, L. Blackburn, A. Broderick, et al. Persistent Non-Gaussian Structure in the Image of Sagittarius A* at 86 GHz. *ApJ*, 915(2):99, July 2021. doi:10.3847/1538-4357/ac00b0.
- [25] Avery E. Broderick, Dominic W. Pesce, **Paul Tiede**, Hung-Yi Pu, et al. Hybrid Very Long Baseline Interferometry Imaging and Modeling with THEMIS. *ApJ*, 898(1):9, July 2020. doi:10.3847/1538-4357/ab9c1f.
- [26] Avery E. Broderick, **Paul Tiede**, Philip Chang, Astrid Lamberts, et al. Missing Gamma-Ray Halos and the Need for New Physics in the Gamma-Ray Sky. *ApJ*, 868(2):87, December 2018. doi:10.3847/1538-4357/aae5f2.
- [27] Avery E. Broderick, **Paul Tiede**, Mohamad Shalaby, Christoph Pfrommer, et al. Bow Ties in the Sky. I: The Angular Structure of Inverse Compton Gamma-Ray Halos in the Fermi Sky. *ApJ*, 832(2):109, December 2016. doi:10.3847/0004-637X/832/2/109.

Collaboration Publications

- [1] Ciriaco Goddi, Douglas F. Carlos, Geoffrey B. Crew, Lynn D. Matthews, et al. First polarization study of the M87 jet and active galactic nuclei at submillimeter wavelengths with ALMA. *Astronomy and Astrophysics*, 699:A265, July 2025. doi:10.1051/0004-6361/202554140.
- [2] Jan Röder, Maciek Wielgus, Andrei P. Lobanov, Thomas P. Krichbaum, et al. A multifrequency study of sub-parsec jets with the Event Horizon Telescope. *Astronomy and Astrophysics*, 695:A233, March 2025. doi:10.1051/0004-6361/202452600.
- [3] J. C. Algaba, M. Baloković, S. Chandra, W. Y. Cheong, et al. Broadband multi-wavelength properties of M87 during the 2018 EHT campaign including a very high energy flaring episode. *Astronomy and Astrophysics*, 692:A140, December 2024. doi:10.1051/0004-6361/202450497.
- [4] Anne-Kathrin Baczko, Matthias Kadler, Eduardo Ros, Christian M. Fromm, et al. The putative center in NGC 1052. *Astronomy and Astrophysics*, 692:A205, December 2024. doi:10.1051/0004-6361/202450898.
- [5] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First Sagittarius A* Event Horizon Telescope Results. VIII. Physical Interpretation of the Polarized Ring. *ApJL*, 964(2):L26, April 2024. doi:10.3847/2041-8213/ad2df1.
- [6] Alexander W. Raymond, Sheperd S. Doeleman, Keiichi Asada, Lindy Blackburn, et al. First Very Long Baseline Interferometry Detections at 870 μm . *AJ*, 168(3):130, September 2024. doi:10.3847/1538-3881/ad5bdb.
- [7] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First M87 Event Horizon Telescope Results. IX. Detection of Near-horizon Circular Polarization. *ApJL*, 957(2):L20, November 2023. doi:10.3847/2041-8213/acff70.
- [8] Svetlana Jorstad, Maciek Wielgus, Rocco Lico, Sara Issaoun, et al. The Event Horizon Telescope Image of the Quasar NRAO 530. *ApJ*, 943(2):170, February 2023. doi:10.3847/1538-4357/aca8a8.
- [9] Ben S. Prather, Jason Dexter, Monika Moscibrodzka, Hung-Yi Pu, et al. Comparison of Polarized Radiative Transfer Codes Used by the EHT Collaboration. *ApJ*, 950(1):35, June 2023. doi:10.3847/1538-4357/acc586.
- [10] Freek Roelofs, Michael D. Johnson, Andrew Chael, Michael Janssen, et al. Polarimetric Geometric Modeling for mm-VLBI Observations of Black Holes. *ApJL*, 957(2):L21, November 2023. doi:10.3847/2041-8213/acff6f.
- [11] Pablo Torne, Kuo Liu, Ralph P. Eatough, Jompoj Wongphechauxsorn, et al. A Search for Pulsars around Sgr A* in the First Event Horizon Telescope Data Set. *ApJ*, 959(1):14, December 2023. doi:10.3847/1538-4357/acf4f2.
- [12] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First Sagittarius A* Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole in the Center of the Milky Way. *ApJL*, 930(2):L12, May 2022. doi:10.3847/2041-8213/ac6674.

- [13] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First Sagittarius A* Event Horizon Telescope Results. II. EHT and Multiwavelength Observations, Data Processing, and Calibration. *ApJl*, 930(2):L13, May 2022. doi:10.3847/2041-8213/ac6675.
- [14] Joseph Farah, Peter Galison, Kazunori Akiyama, Katherine L. Bouman, et al. Selective Dynamical Imaging of Interferometric Data. *ApJl*, 930(2):L18, May 2022. doi:10.3847/2041-8213/ac6615.
- [15] Boris Georgiev, Dominic W. Pesce, Avery E. Broderick, George N. Wong, et al. A Universal Power-law Prescription for Variability from Synthetic Images of Black Hole Accretion Flows. *ApJl*, 930(2):L20, May 2022. doi:10.3847/2041-8213/ac65eb.
- [16] Kaushik Satapathy, Dimitrios Psaltis, Feryal Özel, Lia Medeiros, et al. The Variability of the Black Hole Image in M87 at the Dynamical Timescale. *ApJ*, 925(1):13, January 2022. doi:10.3847/1538-4357/ac332e.
- [17] EHT MWL Science Working Group, J. C. Algaba, J. Anczarski, K. Asada, et al. Broadband Multi-wavelength Properties of M87 during the 2017 Event Horizon Telescope Campaign. *ApJl*, 911(1):L11, April 2021. doi:10.3847/2041-8213/abef71.
- [18] Event Horizon Telescope Collaboration, Kazunori Akiyama, Juan Carlos Algaba, Antxon Alberdi, et al. First M87 Event Horizon Telescope Results. VII. Polarization of the Ring. *ApJl*, 910(1):L12, March 2021. doi:10.3847/2041-8213/abe71d.
- [19] Event Horizon Telescope Collaboration, Kazunori Akiyama, Juan Carlos Algaba, Antxon Alberdi, et al. First M87 Event Horizon Telescope Results. VIII. Magnetic Field Structure near The Event Horizon. *ApJl*, 910(1):L13, March 2021. doi:10.3847/2041-8213/abe4de.
- [20] Ciriaco Goddi, Iván Martí-Vidal, Hugo Messias, Geoffrey C. Bower, et al. Polarimetric Properties of Event Horizon Telescope Targets from ALMA. *ApJl*, 910(1):L14, March 2021. doi:10.3847/2041-8213/abee6a.
- [21] Michael Janssen, Heino Falcke, Matthias Kadler, Eduardo Ros, et al. Event Horizon Telescope observations of the jet launching and collimation in Centaurus A. *Nature Astronomy*, 5:1017–1028, July 2021. doi:10.1038/s41550-021-01417-w.
- [22] Prashant Kocherlakota, Luciano Rezzolla, Heino Falcke, Christian M. Fromm, et al. Constraints on black-hole charges with the 2017 EHT observations of M87*. *PRD*, 103(10):104047, May 2021. doi:10.1103/PhysRevD.103.104047.
- [23] Ramesh Narayan, Daniel C. M. Palumbo, Michael D. Johnson, Zachary Gelles, et al. The Polarized Image of a Synchrotron-emitting Ring of Gas Orbiting a Black Hole. *ApJ*, 912(1):35, May 2021. doi:10.3847/1538-4357/abf117.
- [24] Roman Gold, Avery E. Broderick, Ziri Younsi, Christian M. Fromm, et al. Verification of Radiative Transfer Schemes for the EHT. *ApJ*, 897(2):148, July 2020. doi:10.3847/1538-4357/ab96c6.
- [25] Jae-Young Kim, Thomas P. Krichbaum, Avery E. Broderick, Maciek Wielgus, et al. Event Horizon Telescope imaging of the archetypal blazar 3C 279 at an extreme 20 microarcsecond resolution. *A&A*, 640:A69, August 2020. doi:10.1051/0004-6361/202037493.
- [26] F. Roelofs, M. Janssen, I. Natarajan, R. Deane, et al. SYMBA: An end-to-end VLBI synthetic data generation pipeline. Simulating Event Horizon Telescope observations of M 87. *A&A*, 636:A5, April 2020. doi:10.1051/0004-6361/201936622.
- [27] Maciek Wielgus, Kazunori Akiyama, Lindy Blackburn, Chi-kwan Chan, et al. Monitoring the Morphology of M87* in 2009-2017 with the Event Horizon Telescope. *ApJ*, 901(1):67, September 2020. doi:10.3847/1538-4357/abac0d.
- [28] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole. *ApJl*, 875(1):L1, April 2019. doi:10.3847/2041-8213/ab0ec7.
- [29] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First M87 Event Horizon Telescope Results. II. Array and Instrumentation. *ApJl*, 875(1):L2, April 2019. doi:10.3847/2041-8213/ab0c96.

- [30] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First M87 Event Horizon Telescope Results. III. Data Processing and Calibration. *ApJL*, 875(1):L3, April 2019. doi:10.3847/2041-8213/ab0c57.
- [31] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First M87 Event Horizon Telescope Results. IV. Imaging the Central Supermassive Black Hole. *ApJL*, 875(1):L4, April 2019. doi:10.3847/2041-8213/ab0e85.
- [32] Event Horizon Telescope Collaboration, Kazunori Akiyama, Antxon Alberdi, Walter Alef, et al. First M87 Event Horizon Telescope Results. V. Physical Origin of the Asymmetric Ring. *ApJL*, 875(1):L5, April 2019. doi:10.3847/2041-8213/ab0f43.
- [33] Oliver Porth, Koushik Chatterjee, Ramesh Narayan, Charles F. Gammie, et al. The Event Horizon General Relativistic Magnetohydrodynamic Code Comparison Project. *ApJs*, 243(2):26, August 2019. doi:10.3847/1538-4365/ab29fd.